

# 1D CNN — Architecture Variant Comparison

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## Configuration (full)

Every parameter from the configuration cell is reproduced verbatim below. Copy the values back into the notebook config cell to regenerate this run. Derived counts (trials / channels / classes) are shown in *italics* in the Derived quantities section.

### Data locations & class filter

|                    |                                                               |
|--------------------|---------------------------------------------------------------|
| DATA_PATHS[0]      | /home/jestin/ThesisRepo/ML/NewTrainingData/NewNewDynamic      |
| DATA_ROOT (legacy) | None                                                          |
| BATCH_TEST_ROOT    | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/NewNewDynamic |
| TEST_USER          | 8_Jestin                                                      |
| INCLUDE_LABELS     | None                                                          |
| ReportPostfix      | seed43                                                        |

### Sensor selection

|                      |                                                        |
|----------------------|--------------------------------------------------------|
| Selection mode       | SENSOR_GRID dict (fine-grained per-sensor)             |
| USE_LEFT_HAND        | True                                                   |
| USE_RIGHT_HAND       | True                                                   |
| INCLUDE_SEGMENTS     | ['thumb', 'index', 'middle', 'ring', 'pinky', 'wrist'] |
| SENSORS enabled (n)  | 44                                                     |
| SENSORS disabled (n) | 0                                                      |

### SENSOR\_GRID (✓ enabled, ✗ disabled, — not present)

| Hand / Segment | IMU_mid | IMU_prox | Flex_MCP | Flex_PIP |
|----------------|---------|----------|----------|----------|
| left palm      | ✓       | —        | —        | —        |
| left thumb     | ✓       | ✓        | ✓        | ✓        |
| left index     | ✓       | ✓        | ✓        | ✓        |
| left middle    | ✓       | ✓        | ✓        | ✓        |
| left ring      | ✓       | ✓        | ✓        | ✓        |
| left pinky     | ✓       | ✓        | ✓        | ✓        |
| left wrist     | ✓       | —        | —        | —        |
| right palm     | ✓       | —        | —        | —        |
| right thumb    | ✓       | ✓        | ✓        | ✓        |
| right index    | ✓       | ✓        | ✓        | ✓        |
| right middle   | ✓       | ✓        | ✓        | ✓        |
| right ring     | ✓       | ✓        | ✓        | ✓        |
| right pinky    | ✓       | ✓        | ✓        | ✓        |
| right wrist    | ✓       | —        | —        | —        |

### IMU channel modalities

|           |       |
|-----------|-------|
| USE_YPR   | True  |
| USE_QUAT  | False |
| USE_ACCEL | True  |
| USE_FLEX  | False |

### Preprocessing

|                       |                  |
|-----------------------|------------------|
| RESAMPLE_TO_N_STEPS   | 90               |
| APPLY_BUTTERWORTH     | True             |
| BUTTERWORTH_CUTOFF_HZ | 10.0             |
| BUTTERWORTH_ORDER     | 4                |
| SAMPLING_RATE_HZ      | 30.0             |
| NORMALISATION         | per_trial_zscore |

### Train / test split

|                |      |
|----------------|------|
| TEST_SIZE      | 0.1  |
| RANDOM_STATE   | 41   |
| STRATIFY_SPLIT | True |

### Cross-validation

|                      |      |
|----------------------|------|
| RUN_CROSS_VALIDATION | True |
| CV_FOLDS             | 7    |
| CV_SHUFFLE           | True |

### Augmentation (master switches)

|                       |      |
|-----------------------|------|
| AUGMENT_TRAINING_DATA | True |
|-----------------------|------|

|                                |    |
|--------------------------------|----|
| AUGMENTATION_COPIES_PER_SAMPLE | 6  |
| AUGMENTATION_RANDOM_SEED       | 42 |

AUGMENT\_CONFIG (per transform)

| Transform       | Enabled | apply_prob | Other params                                 |
|-----------------|---------|------------|----------------------------------------------|
| time_shift      | ✓       | 1.0        | max_shift_steps=3, fill_mode=edge            |
| time_warp       | ✓       | 1.0        | speed_range=(0.95, 1.05)                     |
| time_mask       | ✓       | 1.0        | max_masks=2, max_mask_size=3, fill_mode=zero |
| gaussian_noise  | ✗       | 1.0        | std_ratio=0.02                               |
| amplitude_scale | ✓       | 1.0        | scale_range=(0.95, 1.05)                     |
| baseline_offset | ✓       | 1.0        | offset_std_ratio=0.02                        |
| channel_dropout | ✓       | 1.0        | drop_fraction=0.03, fill_mode=zero           |
| left_right_swap | ✗       | 0.5        | —                                            |

Training hyperparameters

|                        |     |
|------------------------|-----|
| TRAIN_EPOCHS           | 60  |
| TRAIN_BATCH_SIZE       | 16  |
| TRAIN_VALIDATION_SPLIT | 0.2 |
| TRAIN_VERBOSE          | 1   |

Early stopping

|                             |          |
|-----------------------------|----------|
| EARLY_STOPPING_ENABLED      | True     |
| EARLY_STOPPING_MONITOR      | val_loss |
| EARLY_STOPPING_PATIENCE     | 8        |
| EARLY_STOPPING_MIN_DELTA    | 0.0      |
| EARLY_STOPPING_RESTORE_BEST | True     |
| EARLY_STOPPING_VERBOSE      | 1        |

Output paths

|                     |                                                                                          |
|---------------------|------------------------------------------------------------------------------------------|
| MODEL_OUTPUT_DIR    | /home/jestin/ThesisRepo/ML/NewReports/NewDynamicReports/NN_Models                        |
| EXPERIMENT_LOG_PATH | /home/jestin/ThesisRepo/ML/NewReports/NewDynamicReports/NN_Models/experiment_results.csv |

Variants in this run

| Name        | Builder           | Description                                                                                                                                                |
|-------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baseline    | build_baseline    | Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax                                                                             |
| Shallow     | build_shallow     | Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax                                                                              |
| Deep        | build_deep        | Three Conv blocks:<br>Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax                                                      |
| BN_GAP      | build_bn_gap      | Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit                                                                              |
| WideKernel  | build_wide_kernel | Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax                                                                             |
| CNN_LSTM    | build_cnn_lstm    | Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax                                                                             |
| CNN_BiLSTM  | build_cnn_bilstm  | Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax                                                                           |
| Model_I_1D  | build_model_i_1d  | 1-D adaptation of published Model I: Conv1D(32,k=5,s=3)→Pool→BN→Conv1D(64,k=2)→Conv1D(64,k=2)→Pool→Flatten→Dense(64)→Drop(0.5)→Softmax                     |
| Model_II_1D | build_model_ii_1d | 1-D adaptation of published Model II (CNN+LSTM): Conv1D(32,k=5,s=3)→Conv1D(32,k=3)→BN→Pool→Drop→LSTM(64,seq)→Drop→LSTM(64)→Drop→Dense(128)→BN→Drop→Softmax |

Derived quantities (read-only — computed from data)

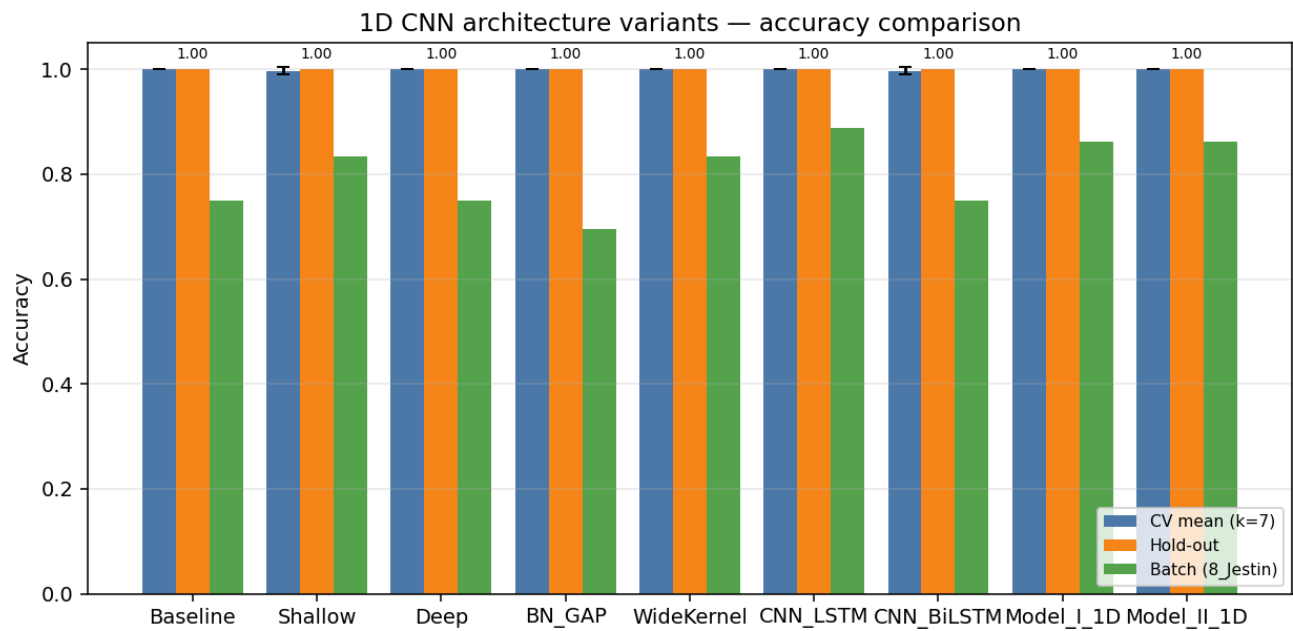
|                           |                                                                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Total trials (X.shape[0]) | 360                                                                                                                                                             |
| Train pool size           | 324                                                                                                                                                             |
| Hold-out size             | 36                                                                                                                                                              |
| Sequence length           | 90                                                                                                                                                              |
| Sensor channels           | 144                                                                                                                                                             |
| Number of classes         | 6                                                                                                                                                               |
| Class names               | Double_ClosedHand_Flip_OpenHand, Double_FingerTips, Double_Okay_Ring_Middle_Pinky, Double_OpenHand_Flip_ClosedHand, Double_OpenHand_LPunch_R_Punch, Double_Snap |
| len(SENSOR_COLS)          | 144                                                                                                                                                             |

Variant comparison (summary)

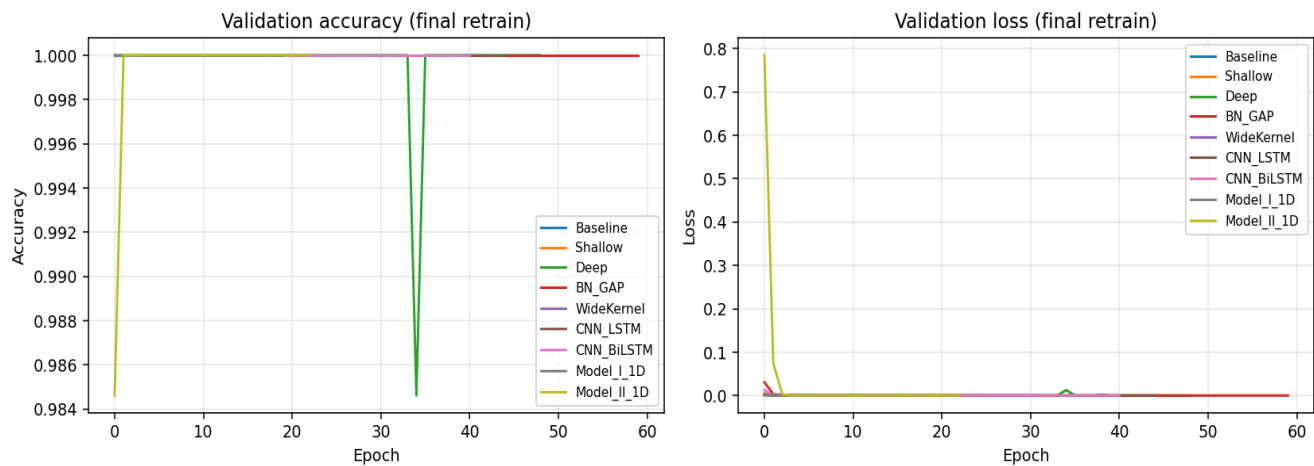
| Variant    | Params  | CV mean (k=7) | CV std | Hold-out acc | Hold-out loss | Batch acc | Batch n |
|------------|---------|---------------|--------|--------------|---------------|-----------|---------|
| Baseline   | 109,094 | 1.0000        | 0.0000 | 1.0000       | 0.0000        | 0.7500    | 27/36   |
| Shallow    | 67,334  | 0.9970        | 0.0074 | 1.0000       | 0.0000        | 0.8333    | 30/36   |
| Deep       | 199,782 | 1.0000        | 0.0000 | 1.0000       | 0.0000        | 0.7500    | 27/36   |
| BN_GAP     | 53,094  | 1.0000        | 0.0000 | 1.0000       | 0.0000        | 0.6944    | 25/36   |
| WideKernel | 123,430 | 1.0000        | 0.0000 | 1.0000       | 0.0000        | 0.8333    | 30/36   |
| CNN_LSTM   | 64,294  | 1.0000        | 0.0000 | 1.0000       | 0.0000        | 0.8889    | 32/36   |
| CNN_BiLSTM | 101,414 | 0.9970        | 0.0074 | 1.0000       | 0.0000        | 0.7500    | 27/36   |
| Model_I_1D | 118,438 | 1.0000        | 0.0000 | 1.0000       | 0.0000        | 0.8611    | 31/36   |

|             |        |        |        |        |        |        |       |
|-------------|--------|--------|--------|--------|--------|--------|-------|
| Model_II_1D | 93,766 | 1.0000 | 0.0000 | 1.0000 | 0.0000 | 0.8611 | 31/36 |
|-------------|--------|--------|--------|--------|--------|--------|-------|

Best on hold-out: **Baseline** · Best on CV: **Baseline**



Training curves (final retrain on full training pool)



# Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 109,094         |
| CV mean ± std            | 1.0000 ± 0.0000 |
| Hold-out test acc / loss | 1.0000 · 0.0000 |
| Batch-test acc           | 0.7500 (27/36)  |

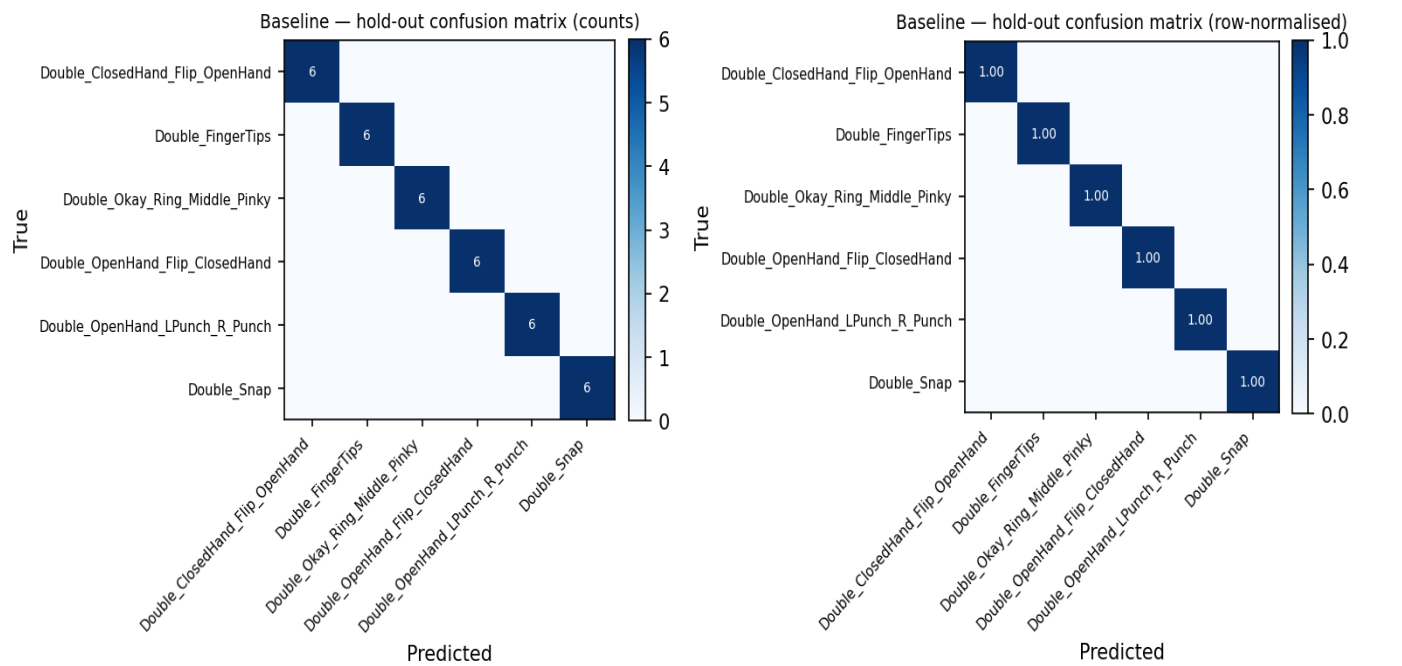
## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 2    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 3    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 4    | 1946             | 46  | 1.0000  | 1.0000   | 0.9783    |
| 5    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 6    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 7    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |

## Per-class hold-out metrics

| Class                           | Precision | Recall | F1    | Support |
|---------------------------------|-----------|--------|-------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_FingerTips               | 1.000     | 1.000  | 1.000 | 6       |
| Double_Okay_Ring_Middle_Pinky   | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_Flip_ClosedHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_LPunch_R_Punch  | 1.000     | 1.000  | 1.000 | 6       |
| Double_Snap                     | 1.000     | 1.000  | 1.000 | 6       |
| macro avg                       | 1.000     | 1.000  | 1.000 | 36      |
| weighted avg                    | 1.000     | 1.000  | 1.000 | 36      |

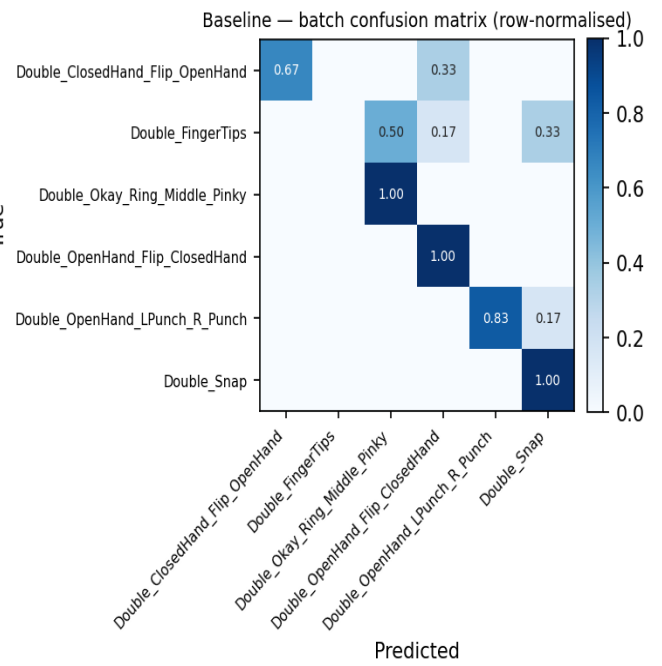
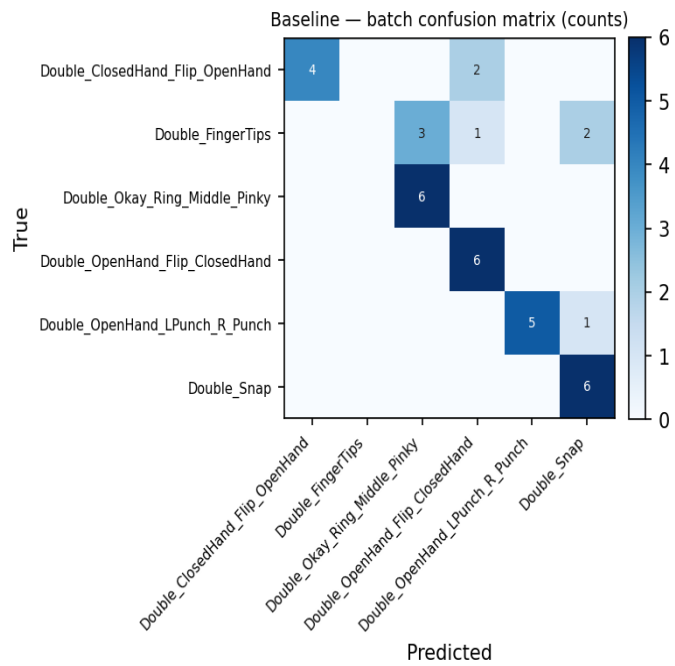
## Hold-out confusion matrix



## Batch test per class (8\_Jestin)

| Class                           | Correct | Total | Accuracy |
|---------------------------------|---------|-------|----------|
| Double_ClosedHand_Flip_OpenHand | 4       | 6     | 0.667    |
| Double_FingerTips               | 0       | 6     | 0.000    |
| Double_Okay_Ring_Middle_Pinky   | 6       | 6     | 1.000    |
| Double_OpenHand_Flip_ClosedHand | 6       | 6     | 1.000    |
| Double_OpenHand_LPunch_R_Punch  | 5       | 6     | 0.833    |
| Double_Snap                     | 6       | 6     | 1.000    |
| Overall                         | 27      | 36    | 0.750    |

## Batch confusion matrix (8\_Jestin)



# Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 67,334          |
| CV mean ± std            | 0.9970 ± 0.0074 |
| Hold-out test acc / loss | 1.0000 · 0.0000 |
| Batch-test acc           | 0.8333 (30/36)  |

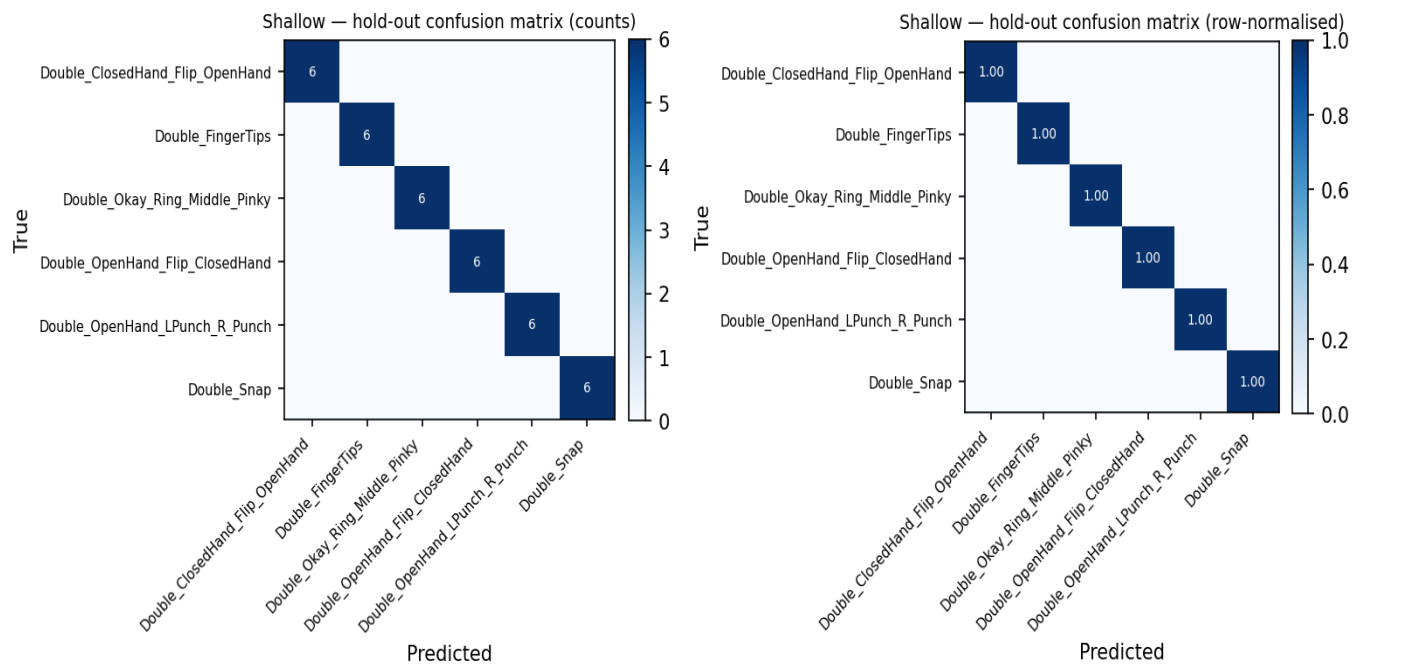
## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1939             | 47  | 0.9787  | 0.9787   | 0.9787    |
| 2    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 3    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 4    | 1946             | 46  | 1.0000  | 1.0000   | 0.9783    |
| 5    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 6    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 7    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |

## Per-class hold-out metrics

| Class                           | Precision | Recall | F1    | Support |
|---------------------------------|-----------|--------|-------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_FingerTips               | 1.000     | 1.000  | 1.000 | 6       |
| Double_Okay_Ring_Middle_Pinky   | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_Flip_ClosedHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_LPunch_R_Punch  | 1.000     | 1.000  | 1.000 | 6       |
| Double_Snap                     | 1.000     | 1.000  | 1.000 | 6       |
| macro avg                       | 1.000     | 1.000  | 1.000 | 36      |
| weighted avg                    | 1.000     | 1.000  | 1.000 | 36      |

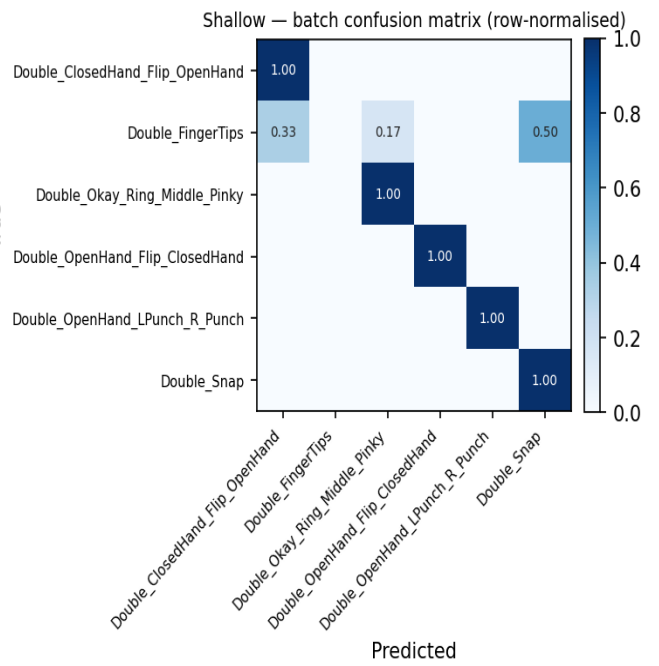
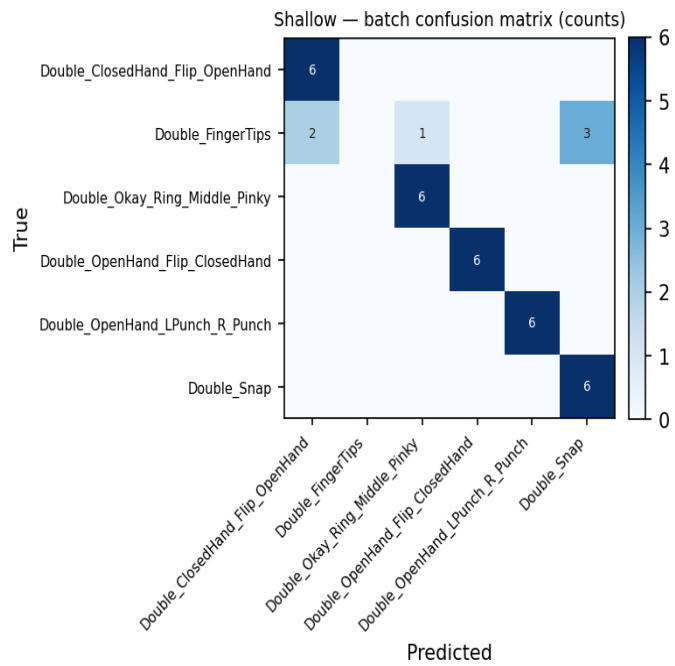
## Hold-out confusion matrix



## Batch test per class (8\_Jestin)

| Class                           | Correct | Total | Accuracy |
|---------------------------------|---------|-------|----------|
| Double_ClosedHand_Flip_OpenHand | 6       | 6     | 1.000    |
| Double_FingerTips               | 0       | 6     | 0.000    |
| Double_Okay_Ring_Middle_Pinky   | 6       | 6     | 1.000    |
| Double_OpenHand_Flip_ClosedHand | 6       | 6     | 1.000    |
| Double_OpenHand_LPunch_R_Punch  | 6       | 6     | 1.000    |
| Double_Snap                     | 6       | 6     | 1.000    |
| Overall                         | 30      | 36    | 0.833    |

## Batch confusion matrix (8\_Jestin)



# Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 199,782         |
| CV mean ± std            | 1.0000 ± 0.0000 |
| Hold-out test acc / loss | 1.0000 · 0.0000 |
| Batch-test acc           | 0.7500 (27/36)  |

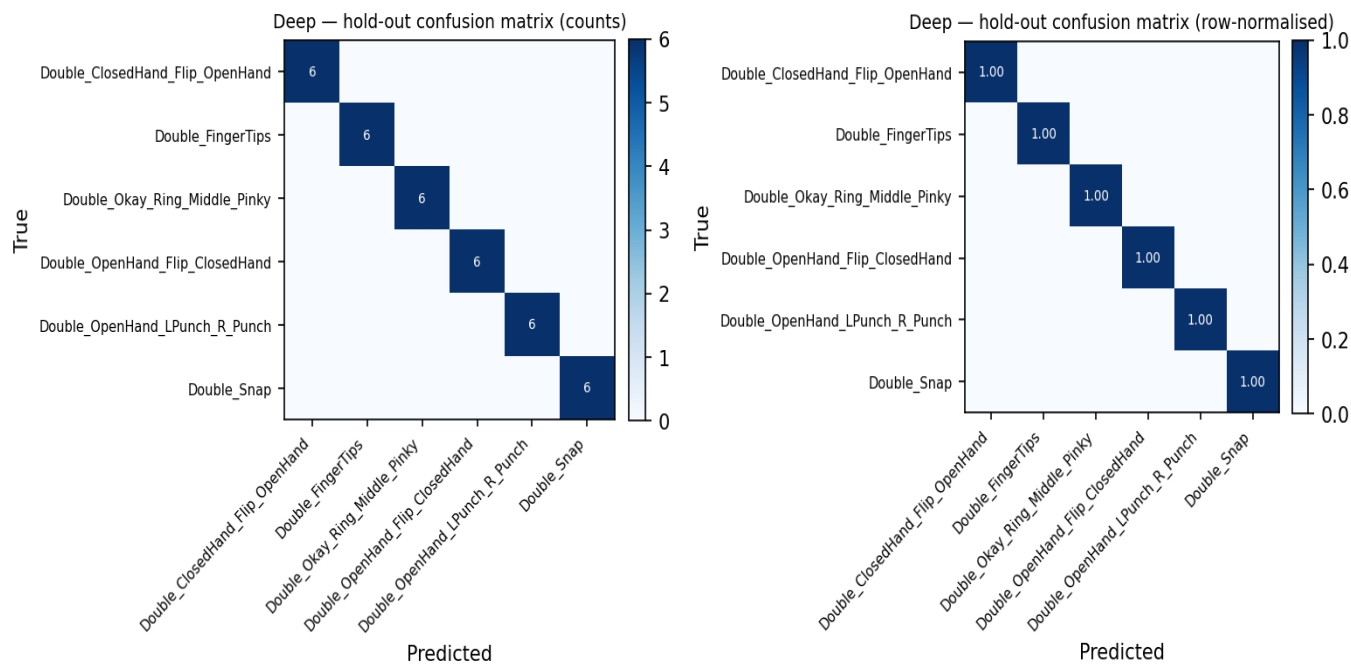
## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 2    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 3    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 4    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 5    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 6    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 7    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |

## Per-class hold-out metrics

| Class                           | Precision | Recall | F1    | Support |
|---------------------------------|-----------|--------|-------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_FingerTips               | 1.000     | 1.000  | 1.000 | 6       |
| Double_Okay_Ring_Middle_Pinky   | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_Flip_ClosedHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_LPunch_R_Punch  | 1.000     | 1.000  | 1.000 | 6       |
| Double_Snap                     | 1.000     | 1.000  | 1.000 | 6       |
| macro avg                       | 1.000     | 1.000  | 1.000 | 36      |
| weighted avg                    | 1.000     | 1.000  | 1.000 | 36      |

## Hold-out confusion matrix

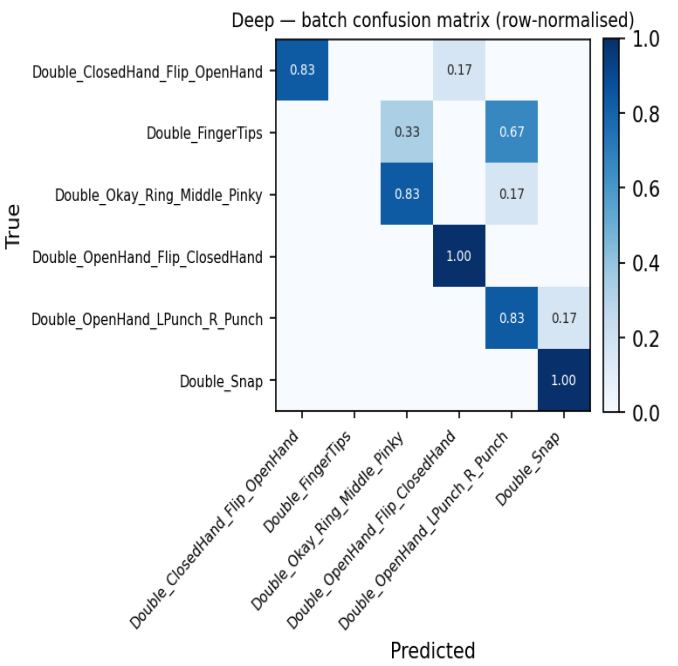
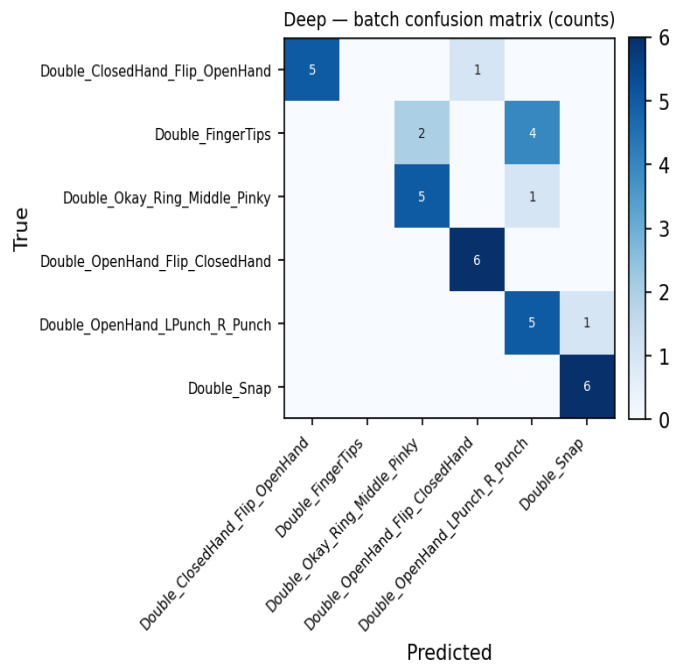


## Batch test per class (8\_Jestin)

| Class                           | Correct | Total | Accuracy |
|---------------------------------|---------|-------|----------|
| Double_ClosedHand_Flip_OpenHand | 5       | 6     | 0.833    |
| Double_FingerTips               | 0       | 6     | 0.000    |
| Double_Okay_Ring_Middle_Pinky   | 5       | 6     | 0.833    |
| Double_OpenHand_Flip_ClosedHand | 6       | 6     | 1.000    |
| Double_OpenHand_LPunch_R_Punch  | 5       | 6     | 0.833    |
| Double_Snap                     | 6       | 6     | 1.000    |
| Overall                         | 27      | 36    | 0.750    |

## Batch confusion matrix (8\_Jestin)





# Variant: BN\_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 53,094          |
| CV mean ± std            | 1.0000 ± 0.0000 |
| Hold-out test acc / loss | 1.0000 · 0.0000 |
| Batch-test acc           | 0.6944 (25/36)  |

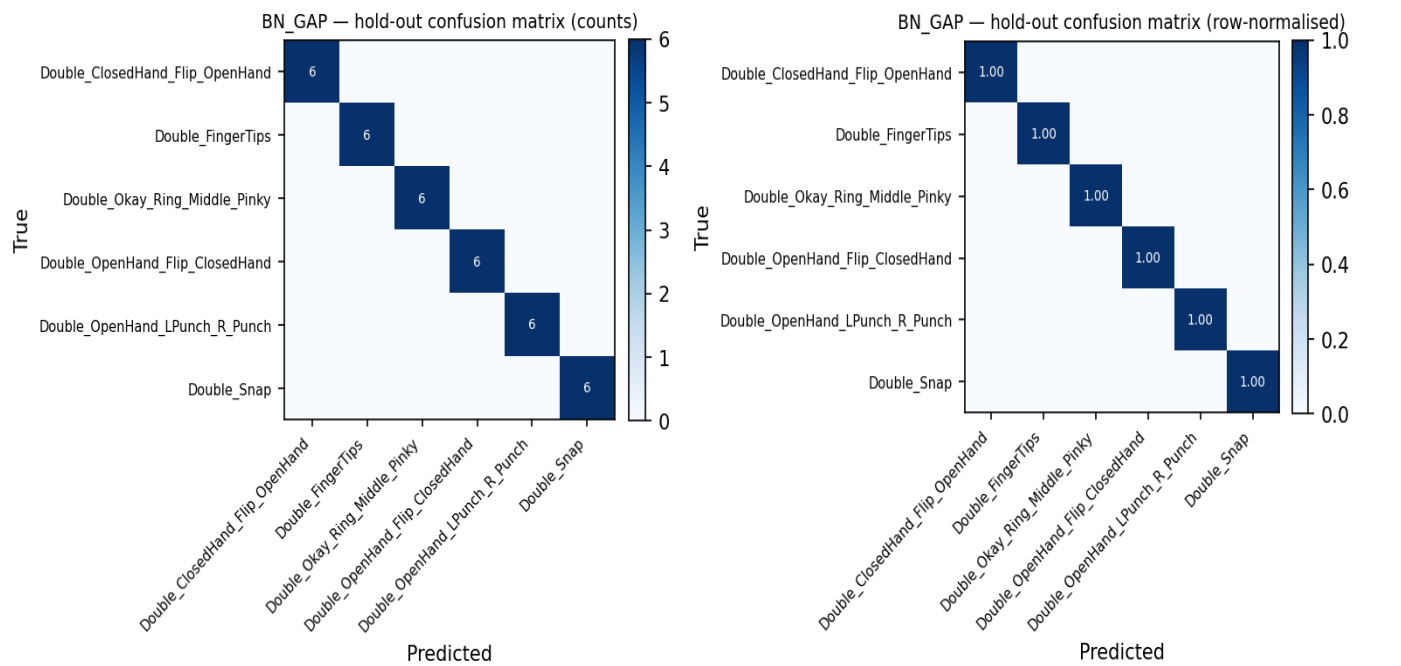
## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 2    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 3    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 4    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 5    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 6    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 7    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |

## Per-class hold-out metrics

| Class                           | Precision | Recall | F1    | Support |
|---------------------------------|-----------|--------|-------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_FingerTips               | 1.000     | 1.000  | 1.000 | 6       |
| Double_Okay_Ring_Middle_Pinky   | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_Flip_ClosedHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_LPunch_R_Punch  | 1.000     | 1.000  | 1.000 | 6       |
| Double_Snap                     | 1.000     | 1.000  | 1.000 | 6       |
| macro avg                       | 1.000     | 1.000  | 1.000 | 36      |
| weighted avg                    | 1.000     | 1.000  | 1.000 | 36      |

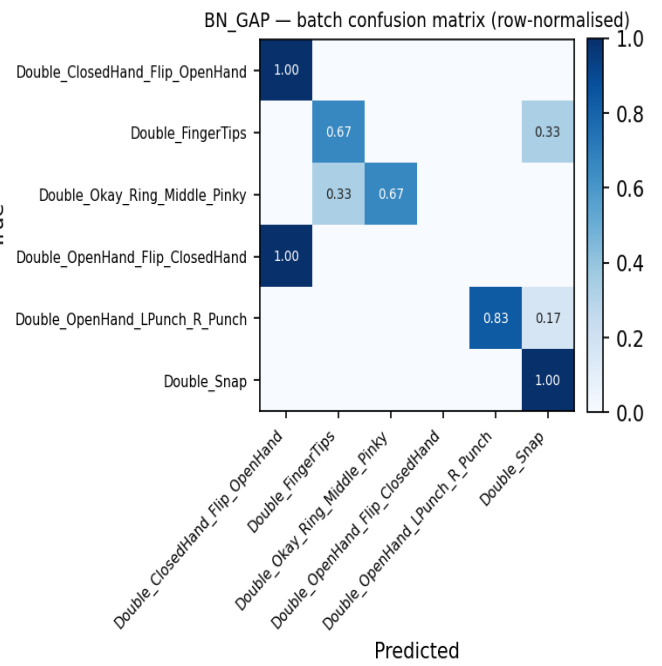
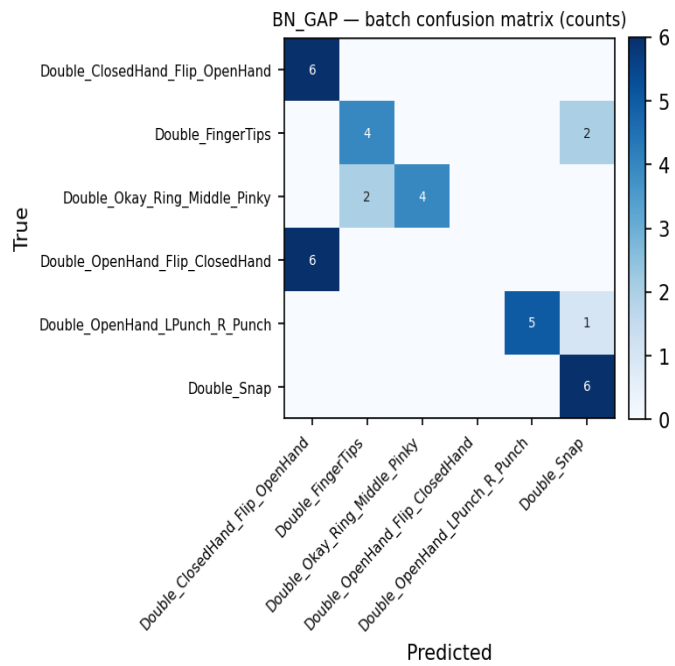
## Hold-out confusion matrix



## Batch test per class (8\_Jestin)

| Class                           | Correct | Total | Accuracy |
|---------------------------------|---------|-------|----------|
| Double_ClosedHand_Flip_OpenHand | 6       | 6     | 1.000    |
| Double_FingerTips               | 4       | 6     | 0.667    |
| Double_Okay_Ring_Middle_Pinky   | 4       | 6     | 0.667    |
| Double_OpenHand_Flip_ClosedHand | 0       | 6     | 0.000    |
| Double_OpenHand_LPunch_R_Punch  | 5       | 6     | 0.833    |
| Double_Snap                     | 6       | 6     | 1.000    |
| Overall                         | 25      | 36    | 0.694    |

## Batch confusion matrix (8\_Jestin)



# Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 123,430         |
| CV mean ± std            | 1.0000 ± 0.0000 |
| Hold-out test acc / loss | 1.0000 · 0.0000 |
| Batch-test acc           | 0.8333 (30/36)  |

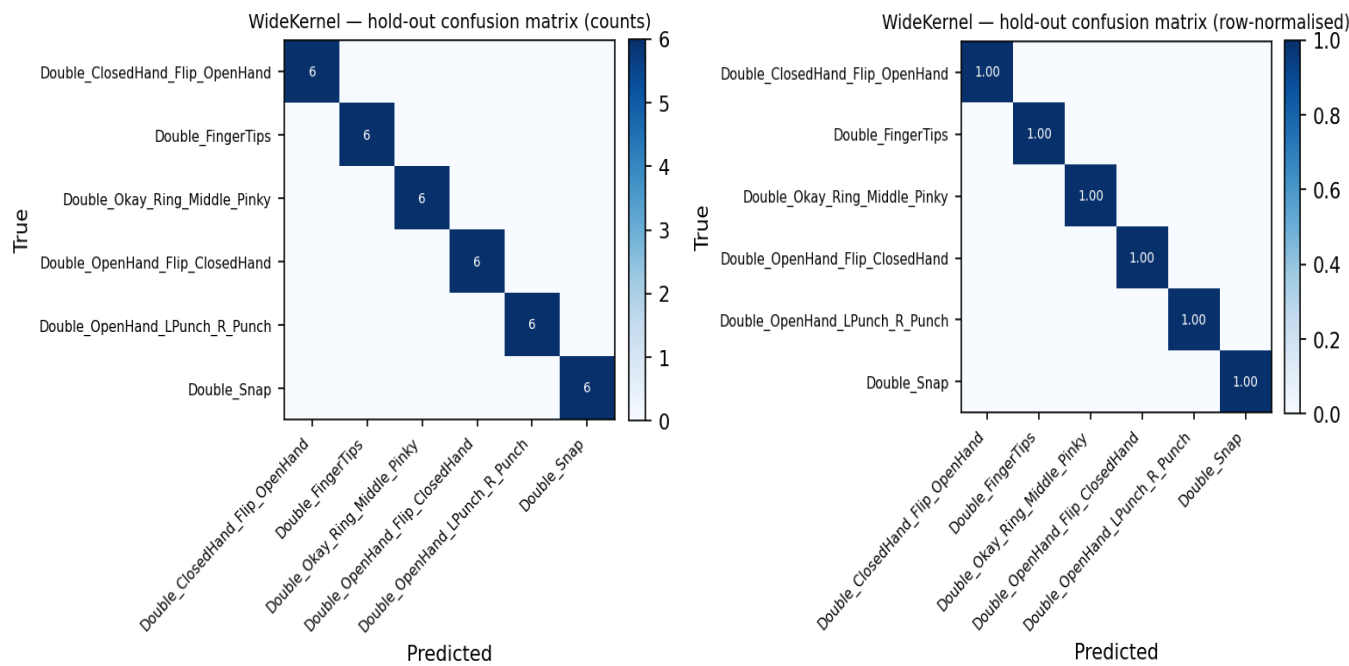
## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 2    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 3    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 4    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 5    | 1946             | 46  | 1.0000  | 1.0000   | 0.9783    |
| 6    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 7    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |

## Per-class hold-out metrics

| Class                           | Precision | Recall | F1    | Support |
|---------------------------------|-----------|--------|-------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_FingerTips               | 1.000     | 1.000  | 1.000 | 6       |
| Double_Okay_Ring_Middle_Pinky   | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_Flip_ClosedHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_LPunch_R_Punch  | 1.000     | 1.000  | 1.000 | 6       |
| Double_Snap                     | 1.000     | 1.000  | 1.000 | 6       |
| macro avg                       | 1.000     | 1.000  | 1.000 | 36      |
| weighted avg                    | 1.000     | 1.000  | 1.000 | 36      |

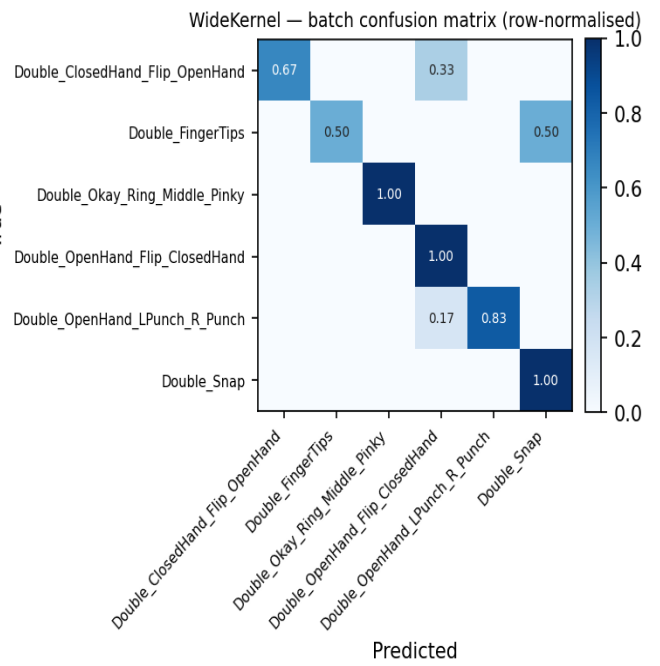
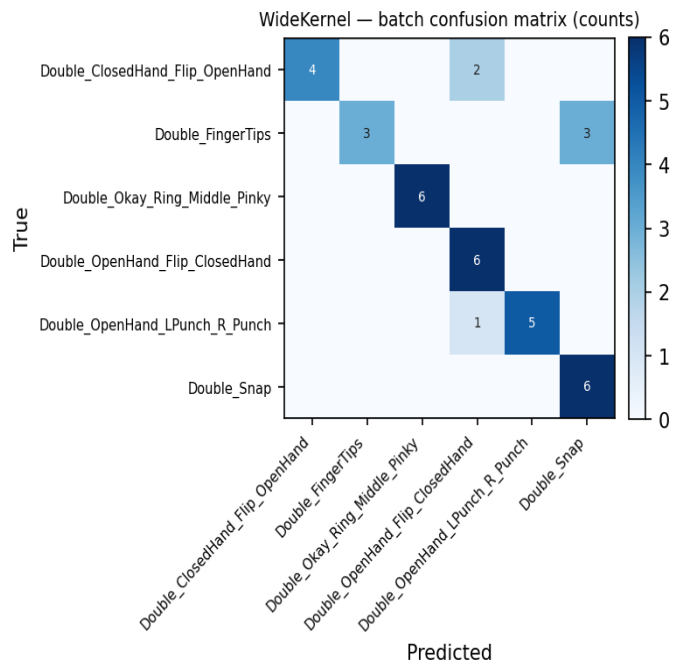
## Hold-out confusion matrix



## Batch test per class (8\_Jestin)

| Class                           | Correct | Total | Accuracy |
|---------------------------------|---------|-------|----------|
| Double_ClosedHand_Flip_OpenHand | 4       | 6     | 0.667    |
| Double_FingerTips               | 3       | 6     | 0.500    |
| Double_Okay_Ring_Middle_Pinky   | 6       | 6     | 1.000    |
| Double_OpenHand_Flip_ClosedHand | 6       | 6     | 1.000    |
| Double_OpenHand_LPunch_R_Punch  | 5       | 6     | 0.833    |
| Double_Snap                     | 6       | 6     | 1.000    |
| Overall                         | 30      | 36    | 0.833    |

## Batch confusion matrix (8\_Jestin)



# Variant: CNN\_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 64,294          |
| CV mean ± std            | 1.0000 ± 0.0000 |
| Hold-out test acc / loss | 1.0000 · 0.0000 |
| Batch-test acc           | 0.8889 (32/36)  |

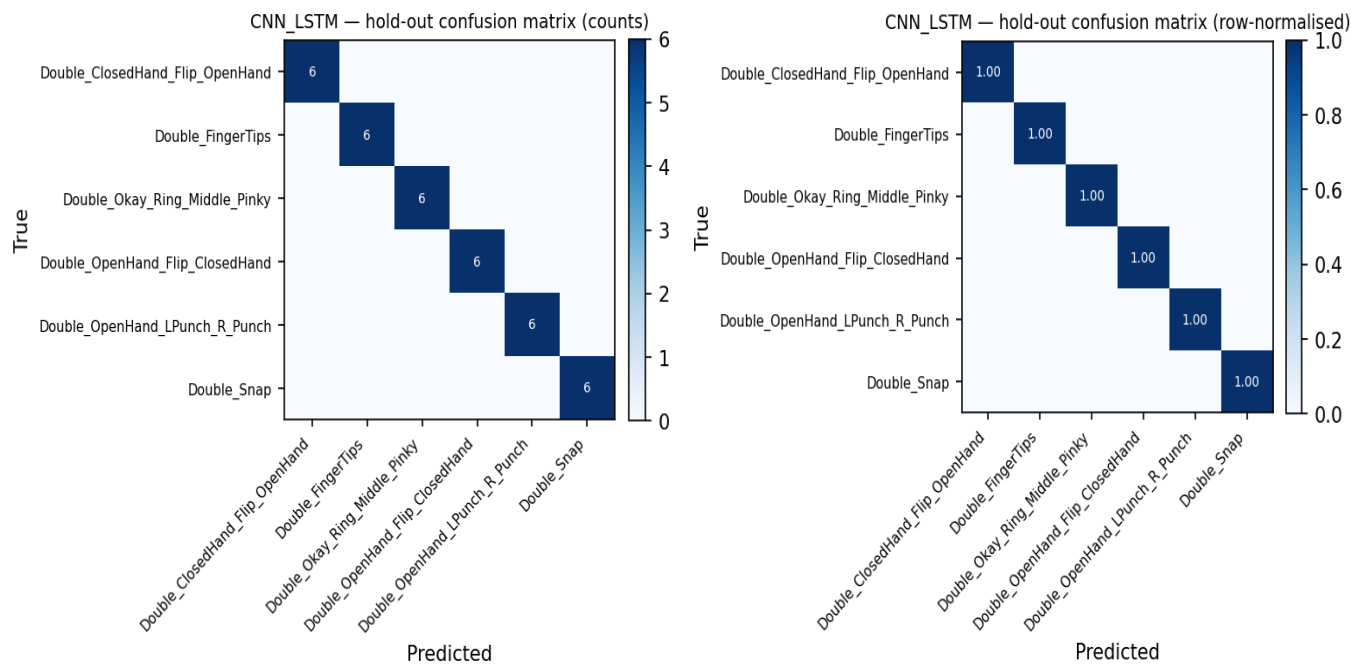
## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 2    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 3    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 4    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 5    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 6    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 7    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |

## Per-class hold-out metrics

| Class                           | Precision | Recall | F1    | Support |
|---------------------------------|-----------|--------|-------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_FingerTips               | 1.000     | 1.000  | 1.000 | 6       |
| Double_Okay_Ring_Middle_Pinky   | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_Flip_ClosedHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_LPunch_R_Punch  | 1.000     | 1.000  | 1.000 | 6       |
| Double_Snap                     | 1.000     | 1.000  | 1.000 | 6       |
| macro avg                       | 1.000     | 1.000  | 1.000 | 36      |
| weighted avg                    | 1.000     | 1.000  | 1.000 | 36      |

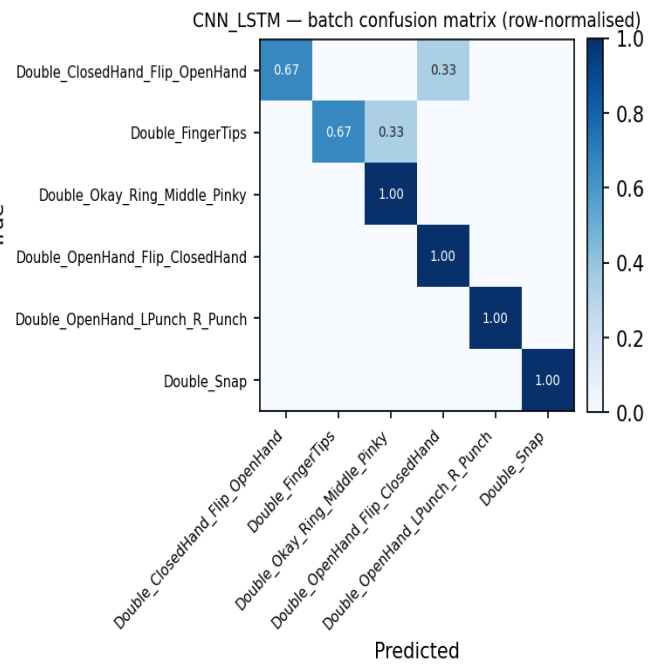
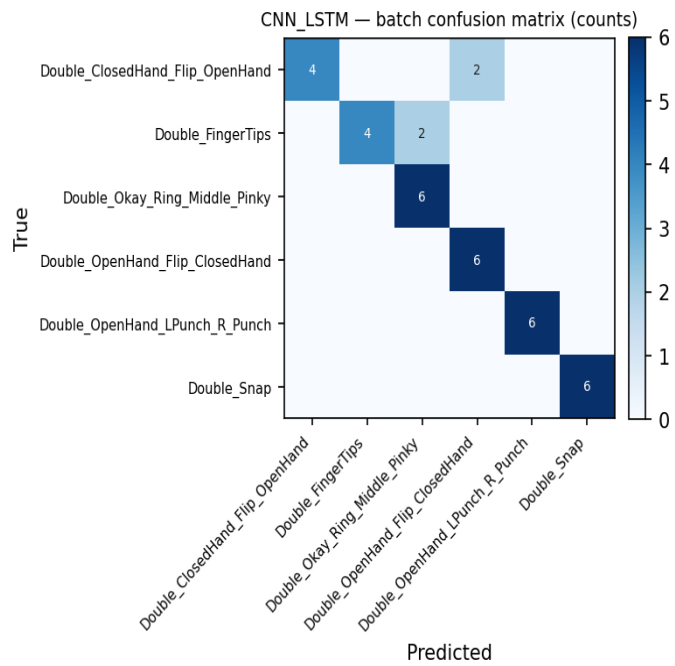
## Hold-out confusion matrix



## Batch test per class (8\_Jestin)

| Class                           | Correct | Total | Accuracy |
|---------------------------------|---------|-------|----------|
| Double_ClosedHand_Flip_OpenHand | 4       | 6     | 0.667    |
| Double_FingerTips               | 4       | 6     | 0.667    |
| Double_Okay_Ring_Middle_Pinky   | 6       | 6     | 1.000    |
| Double_OpenHand_Flip_ClosedHand | 6       | 6     | 1.000    |
| Double_OpenHand_LPunch_R_Punch  | 6       | 6     | 1.000    |
| Double_Snap                     | 6       | 6     | 1.000    |
| Overall                         | 32      | 36    | 0.889    |

## Batch confusion matrix (8\_Jestin)



# Variant: CNN\_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 101,414         |
| CV mean ± std            | 0.9970 ± 0.0074 |
| Hold-out test acc / loss | 1.0000 · 0.0000 |
| Batch-test acc           | 0.7500 (27/36)  |

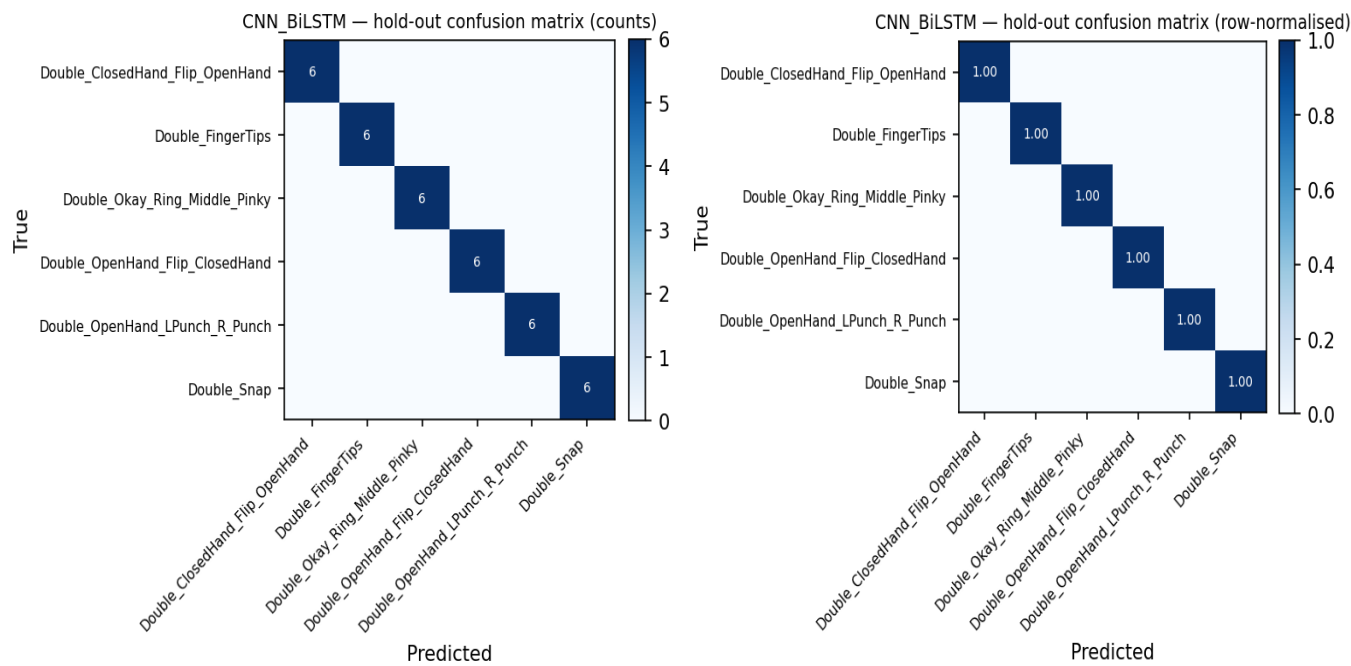
## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1939             | 47  | 0.9787  | 0.9787   | 0.9787    |
| 2    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 3    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 4    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 5    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 6    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 7    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |

## Per-class hold-out metrics

| Class                           | Precision | Recall | F1    | Support |
|---------------------------------|-----------|--------|-------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_FingerTips               | 1.000     | 1.000  | 1.000 | 6       |
| Double_Okay_Ring_Middle_Pinky   | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_Flip_ClosedHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_LPunch_R_Punch  | 1.000     | 1.000  | 1.000 | 6       |
| Double_Snap                     | 1.000     | 1.000  | 1.000 | 6       |
| macro avg                       | 1.000     | 1.000  | 1.000 | 36      |
| weighted avg                    | 1.000     | 1.000  | 1.000 | 36      |

## Hold-out confusion matrix

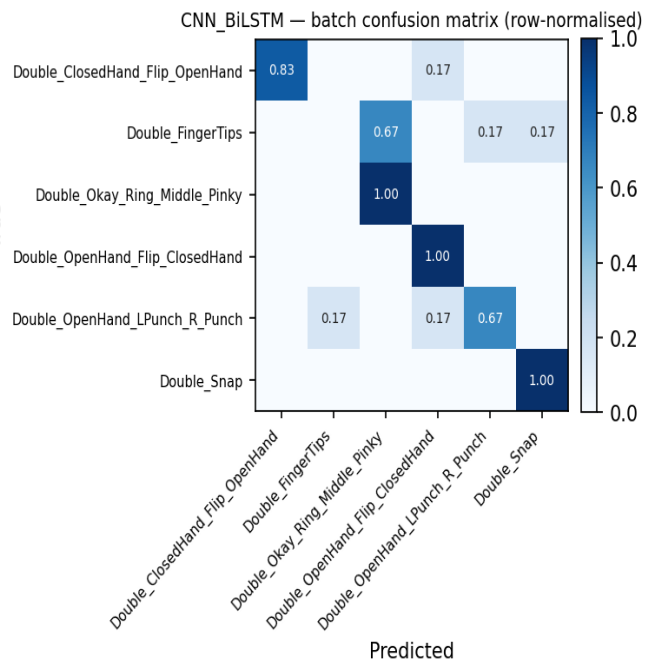
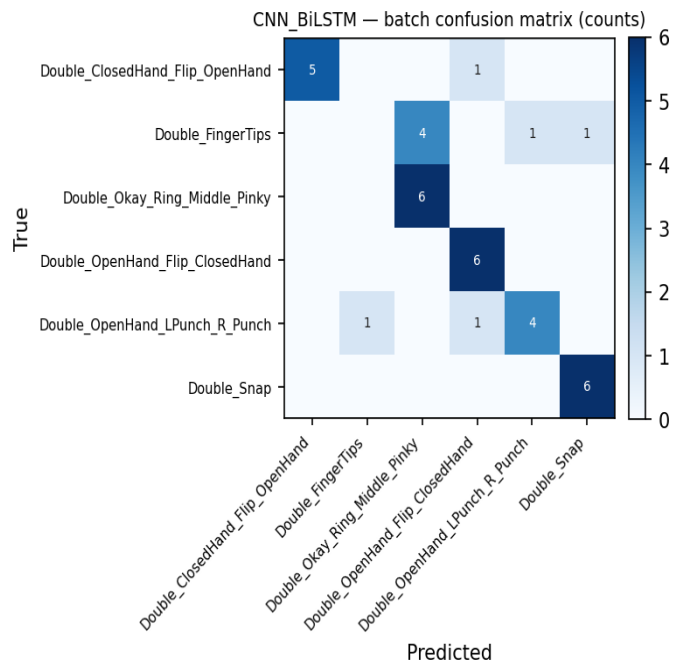


## Batch test per class (8\_Jestin)

| Class                           | Correct | Total | Accuracy |
|---------------------------------|---------|-------|----------|
| Double_ClosedHand_Flip_OpenHand | 5       | 6     | 0.833    |
| Double_FingerTips               | 0       | 6     | 0.000    |
| Double_Okay_Ring_Middle_Pinky   | 6       | 6     | 1.000    |
| Double_OpenHand_Flip_ClosedHand | 6       | 6     | 1.000    |
| Double_OpenHand_LPunch_R_Punch  | 4       | 6     | 0.667    |
| Double_Snap                     | 6       | 6     | 1.000    |
| Overall                         | 27      | 36    | 0.750    |

## Batch confusion matrix (8\_Jestin)





# Variant: Model\_I\_1D

1-D adaptation of published Model I:  
Conv1D(32,k=5,s=3)→Pool→BN→Conv1D(64,k=2)→Conv1D(64,k=2)→Pool→Flatten→Dense(64)→Drop(0.5)→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 118,438         |
| CV mean ± std            | 1.0000 ± 0.0000 |
| Hold-out test acc / loss | 1.0000 · 0.0000 |
| Batch-test acc           | 0.8611 (31/36)  |

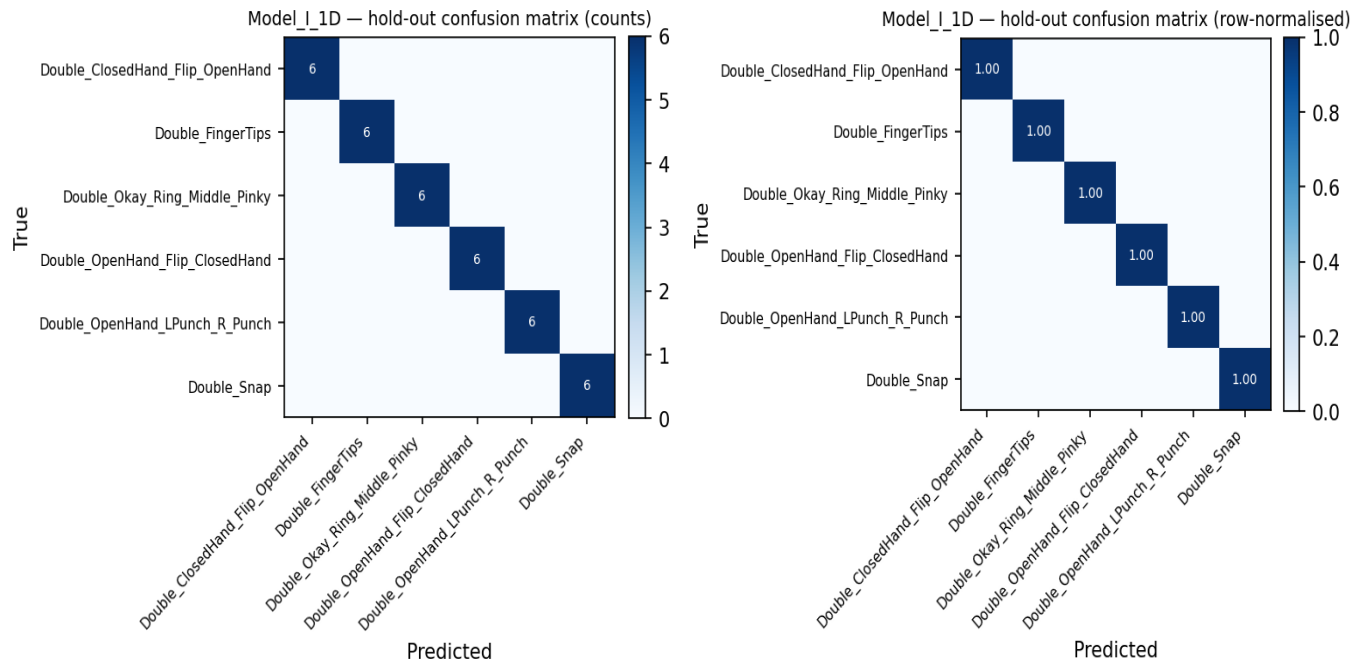
## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1939             | 47  | 1.0000  | 1.0000   | 0.9787    |
| 2    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 3    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 4    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 5    | 1946             | 46  | 1.0000  | 1.0000   | 0.9783    |
| 6    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 7    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |

## Per-class hold-out metrics

| Class                           | Precision | Recall | F1    | Support |
|---------------------------------|-----------|--------|-------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_FingerTips               | 1.000     | 1.000  | 1.000 | 6       |
| Double_Okay_Ring_Middle_Pinky   | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_Flip_ClosedHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_LPunch_R_Punch  | 1.000     | 1.000  | 1.000 | 6       |
| Double_Snap                     | 1.000     | 1.000  | 1.000 | 6       |
| macro avg                       | 1.000     | 1.000  | 1.000 | 36      |
| weighted avg                    | 1.000     | 1.000  | 1.000 | 36      |

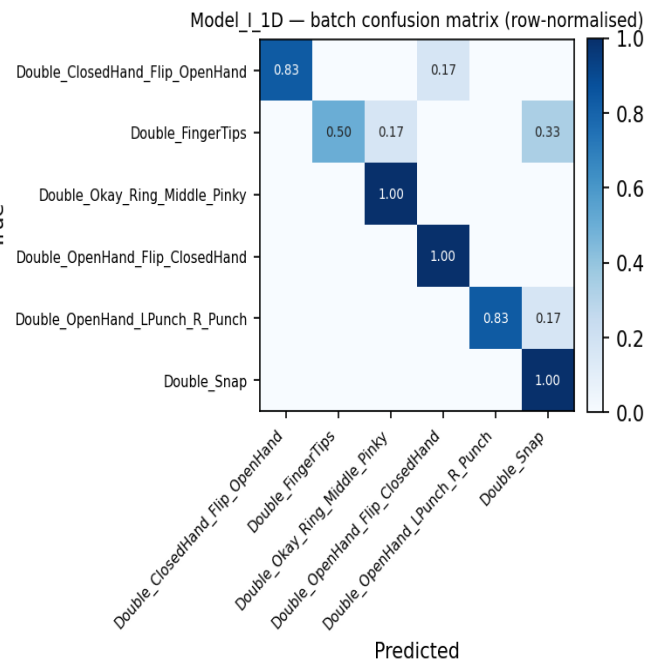
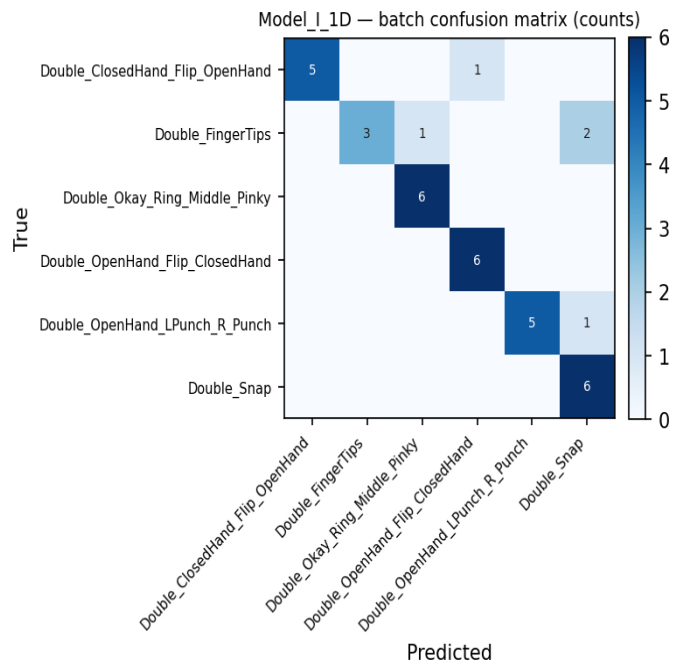
## Hold-out confusion matrix



## Batch test per class (8\_Jestin)

| Class                           | Correct | Total | Accuracy |
|---------------------------------|---------|-------|----------|
| Double_ClosedHand_Flip_OpenHand | 5       | 6     | 0.833    |
| Double_FingerTips               | 3       | 6     | 0.500    |
| Double_Okay_Ring_Middle_Pinky   | 6       | 6     | 1.000    |
| Double_OpenHand_Flip_ClosedHand | 6       | 6     | 1.000    |
| Double_OpenHand_LPunch_R_Punch  | 5       | 6     | 0.833    |
| Double_Snap                     | 6       | 6     | 1.000    |
| Overall                         | 31      | 36    | 0.861    |

## Batch confusion matrix (8\_Jestin)



# Variant: Model\_II\_1D

1-D adaptation of published Model II (CNN+LSTM):  
Conv1D(32,k=5,s=3)→Conv1D(32,k=3)→BN→Pool→Drop→LSTM(64,seq)→Drop→LSTM(64)→Drop→Dense(128)→BN→Drop→Softmax

|                          |                 |
|--------------------------|-----------------|
| Trainable params         | 93,766          |
| CV mean ± std            | 1.0000 ± 0.0000 |
| Hold-out test acc / loss | 1.0000 · 0.0000 |
| Batch-test acc           | 0.8611 (31/36)  |

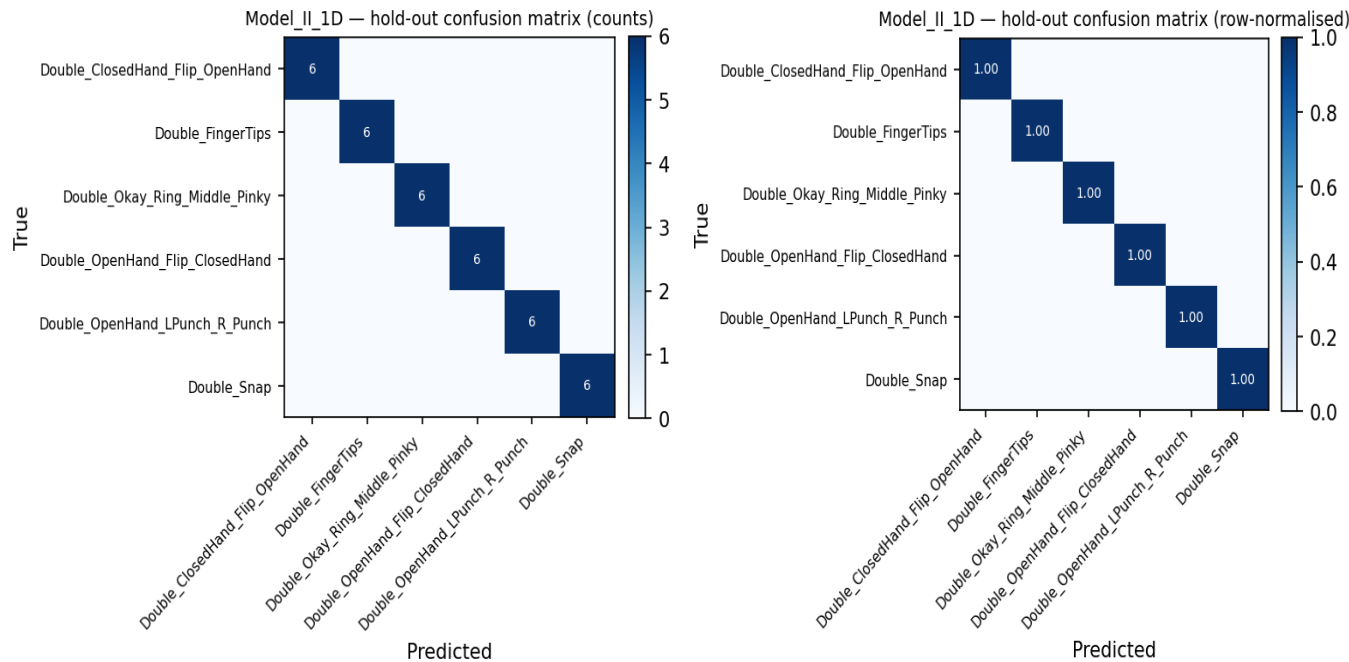
## Cross-validation folds

| Fold | Train (post-aug) | Val | Val acc | Best val | Final val |
|------|------------------|-----|---------|----------|-----------|
| 1    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 2    | 1939             | 47  | 1.0000  | 1.0000   | 1.0000    |
| 3    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 4    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 5    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 6    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |
| 7    | 1946             | 46  | 1.0000  | 1.0000   | 1.0000    |

## Per-class hold-out metrics

| Class                           | Precision | Recall | F1    | Support |
|---------------------------------|-----------|--------|-------|---------|
| Double_ClosedHand_Flip_OpenHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_FingerTips               | 1.000     | 1.000  | 1.000 | 6       |
| Double_Okay_Ring_Middle_Pinky   | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_Flip_ClosedHand | 1.000     | 1.000  | 1.000 | 6       |
| Double_OpenHand_LPunch_R_Punch  | 1.000     | 1.000  | 1.000 | 6       |
| Double_Snap                     | 1.000     | 1.000  | 1.000 | 6       |
| macro avg                       | 1.000     | 1.000  | 1.000 | 36      |
| weighted avg                    | 1.000     | 1.000  | 1.000 | 36      |

## Hold-out confusion matrix



## Batch test per class (8\_Jestin)

| Class                           | Correct | Total | Accuracy |
|---------------------------------|---------|-------|----------|
| Double_ClosedHand_Flip_OpenHand | 6       | 6     | 1.000    |
| Double_FingerTips               | 2       | 6     | 0.333    |
| Double_Okay_Ring_Middle_Pinky   | 6       | 6     | 1.000    |
| Double_OpenHand_Flip_ClosedHand | 6       | 6     | 1.000    |
| Double_OpenHand_LPunch_R_Punch  | 5       | 6     | 0.833    |
| Double_Snap                     | 6       | 6     | 1.000    |
| Overall                         | 31      | 36    | 0.861    |

## Batch confusion matrix (8\_Jestin)

